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Assignment 4
Due Date: 20 April 2026

26/26

Database Development and Design

1.

(13)

a) 3 Ways of implementing the conceptual model

Option1:
Creating one table for the superclass
This table would have:
- Its attributes and all the attributes from both the subclasses
- A subtype discriminator (empType) to distinguish between the two types of employees
- In an instance where the Employee is a Doctor, the attributes of the Nurse will be null.
- Attributes: empNo[pk], empFname, empLname, empDoB, empType, position, shiftHours

Option 2:
Implementing all three tables and having subclasses associated with the super class through appropriate 1-1 relationships.
- Each subtype would have unique attributes and a foreign key from the parent class, which would be shown as [pk][fk] in the design.
- A subtype discriminator (empType) to distinguish between the two types of employees ✓
- Employee: empNo[pk], empFname, empLname, empDoB, empType
- Doctor: empNo[pk][fk], position
- Nurse: empNo[pk][fk], shiftHours

Option 3:
Implementing only the subtype classes
- Each subtype must have all the attributes of the supertype class and its unique attributes.
- The primary key will remain the same as the one in the supertype class
- Doctor: empNo[pk], empFname, empLname, empDoB, position
- Nurse: empNo[pk], empFname, empLname, empDoB, shiftHours.

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b)

For the current situation, implementing only the subtype tables is best (Our option 3 above).

In database modelling, a **mandatory completeness constraint** implies that every instance of the supertype (EMPLOYEE) **must** belong to at least one of the subtypes (DOCTOR or NURSE). Which also implies the EMPLOYEE class is **abstract**.

This means we would only have 2 classes, DOCTOR and NURSE, and would have all attributes from EMPLOYEE as well as their unique attributes.

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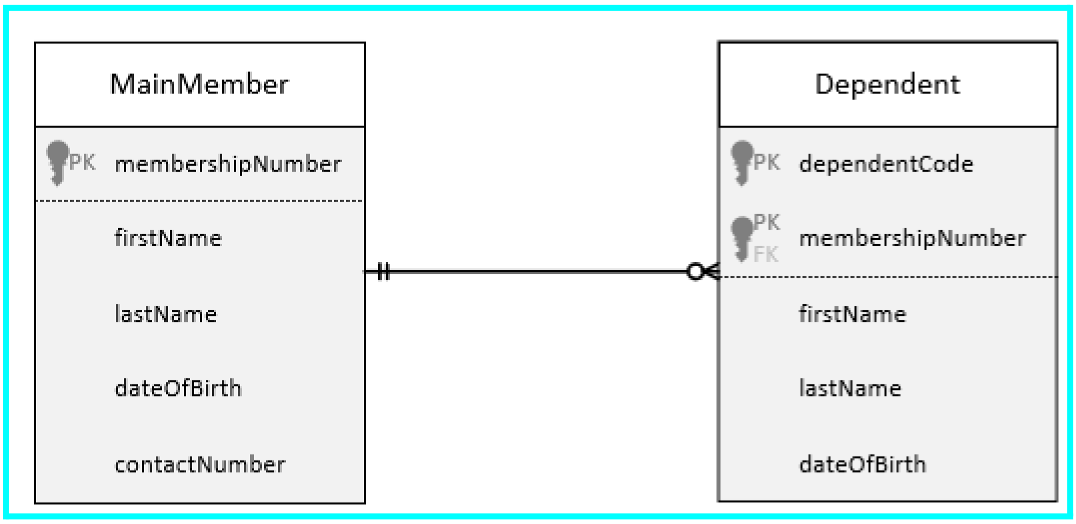
DOCTOR (empNo[pk], empFname, empLname, empDoB, position)
NURSE (empNo[pk], empFname, empLname, empDoB, shiftHours)

2.

a)

[13]



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b) Relational Schema

- MAINMEMBER (membershipNumber[pk], firstName,lastName, dateOfbirth, contactNumber)
- DEPENDENT (dependentCode[pk],membershipNumber[pk][fk], firstName,lastName, dateOfbirth)

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